

Correlation Coefficient

Correlation coefficient is defined as

$$r_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$\text{Cov}(x, y)$: is the covariance between two variables x & y .

This correlation coefficient r_{xy} between two variables x & y represents linear relation between two variable.

$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{n} \sum_{i=1}^n [x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}]$$

$$= \frac{1}{n} \sum_{i=1}^n x_i y_i -$$

$$= \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \frac{1}{n} \sum_{i=1}^n y_i - \bar{y} \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{1}{n} \sum_{i=1}^n x_i y_i - 2 \bar{x} \bar{y} + \bar{x} \bar{y}$$

$$= \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y} = \frac{1}{n} \sum x_i y_i - \left(\frac{\sum x_i}{n} \right) \left(\frac{\sum y_i}{n} \right)$$

Appointments

$$= \frac{1}{n} \sum x_i y_i - \left(\frac{\sum x_i}{n} \right) \left(\frac{\sum y_i}{n} \right)$$

$$\sigma_x = \sqrt{\sum \frac{x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2}, \quad \sigma_y = \sqrt{\sum \frac{y_i^2}{n} - \left(\frac{\sum y_i}{n} \right)^2}$$

Properties of Correlation Coefficient.

Phones to make

① Correlation coefficient is independent (or invariant) under scale change and origin change.

If $u_i = \frac{x_i - h}{k}$, $v_i = \frac{y_i - h'}{k'}$, h, h', k, k' are

constants $\Rightarrow x_i = k u_i + h$, $y_i = k' v_i + h'$

Remarks

$$\Rightarrow \bar{x} = k \bar{u} + h, \quad \bar{y} = k' \bar{v} + h'$$

$$\therefore \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum (k u_i - k \bar{u})(k' v_i - k' \bar{v}) = k k' \text{Cov}(u, v)$$

$$\sigma_x^2 = k^2 \sigma_u^2, \quad \sigma_y^2 = k'^2 \sigma_v^2 \Rightarrow r_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$= \frac{k k' \text{Cov}(u, v)}{k k' \sigma_u \sigma_v} = \frac{\text{Cov}(u, v)}{\sigma_u \sigma_v} = r_{uv}$$

②

$$-1 \leq r_{xy} \leq 1.$$

$$u_i = \frac{x_i - \bar{x}}{\sigma_x}, \quad v_i = \frac{y_i - \bar{y}}{\sigma_y}.$$

$$\sum u_i^2 = \sum \frac{(x_i - \bar{x})^2}{\sigma_x^2} = n \frac{\sigma_x^2}{\sigma_x^2} = n.$$

$$\sum v_i^2 = n.$$

$$\therefore \sum u_i v_i = \sum \frac{(x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y} = \frac{n \text{Cov}(x, y)}{\sigma_x \sigma_y} = n r_{xy}.$$

$$\text{Now } (u_i + v_i)^2 \geq 0.$$

$$\therefore \sum (u_i + v_i)^2 \geq 0.$$

$$\Rightarrow \sum u_i^2 + \sum v_i^2 + 2 \sum u_i v_i \geq 0.$$

$$\Rightarrow n + n + 2n r_{xy} \geq 0$$

$$\Rightarrow 2n(1 + r_{xy}) \geq 0.$$

$$\Rightarrow 1 + r_{xy} \geq 0$$

$$\Rightarrow r_{xy} \geq -1.$$

①

$$(u_i - v_i)^2 \geq 0 \Rightarrow \sum (u_i - v_i)^2 \geq 0.$$

②

$$\Rightarrow 1 - r_{xy} \geq 0 \Rightarrow 1 \geq r_{xy}.$$

$$\text{① \& ②} \Rightarrow -1 \leq r \leq 1.$$

Remarks

S M T W T F S S M T W T F S S M T W T F S S M T
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

JULY

mp.

③

4 v.u. - Tiled. Constant.

$$r(ax+by+cz+d) = \frac{ae}{|ace|} v_{xy}.$$

JUNE

F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	•
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	•

$$\text{var}(X \pm Y) = \sigma_x^2 + \sigma_y^2 \pm 2\text{cov}(X, Y)$$